

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Q1. Comparative Analysis of Star and Bus Topologies

Introduction

Network topology refers to the physical or logical arrangement of devices and communication links in a computer network. **Star topology** and **Bus topology** are two commonly studied network topologies with different structures, costs, reliability levels, and performance characteristics.

In a **Star topology**, every network device is connected individually to a central networking device such as a **switch or hub**. In contrast, a **Bus topology** connects all devices to a single shared **backbone cable**. Therefore, the two topologies differ significantly in terms of cost, reliability, performance, installation, scalability, and maintenance.

1. Structural Comparison

Star Topology

In Star topology, each computer or network device has a separate connection to a central switch or hub. All communication between devices generally passes through this central device.

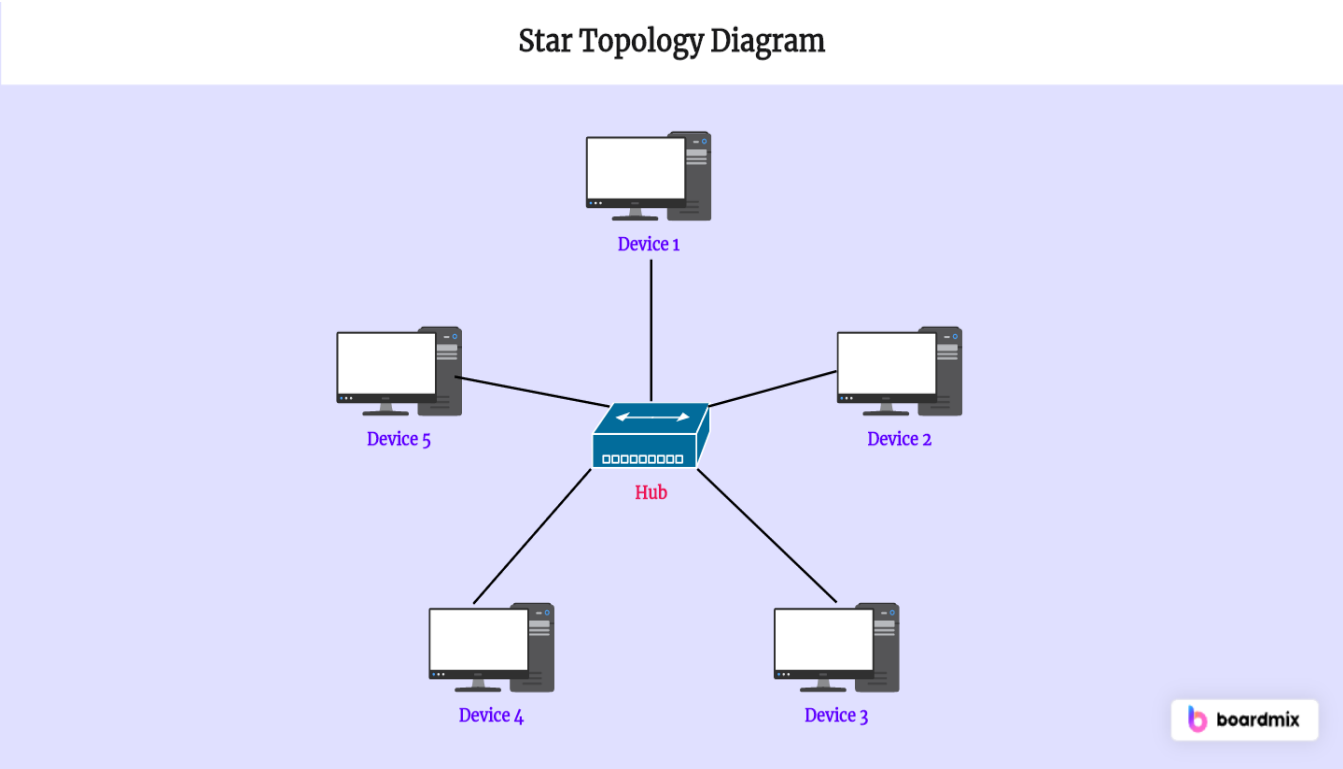


Figure 1: Star Topology Structure

Bus Topology

In Bus topology, all devices share a common backbone cable. Data transmitted by one device travels along the shared medium, and the intended device receives the relevant data.

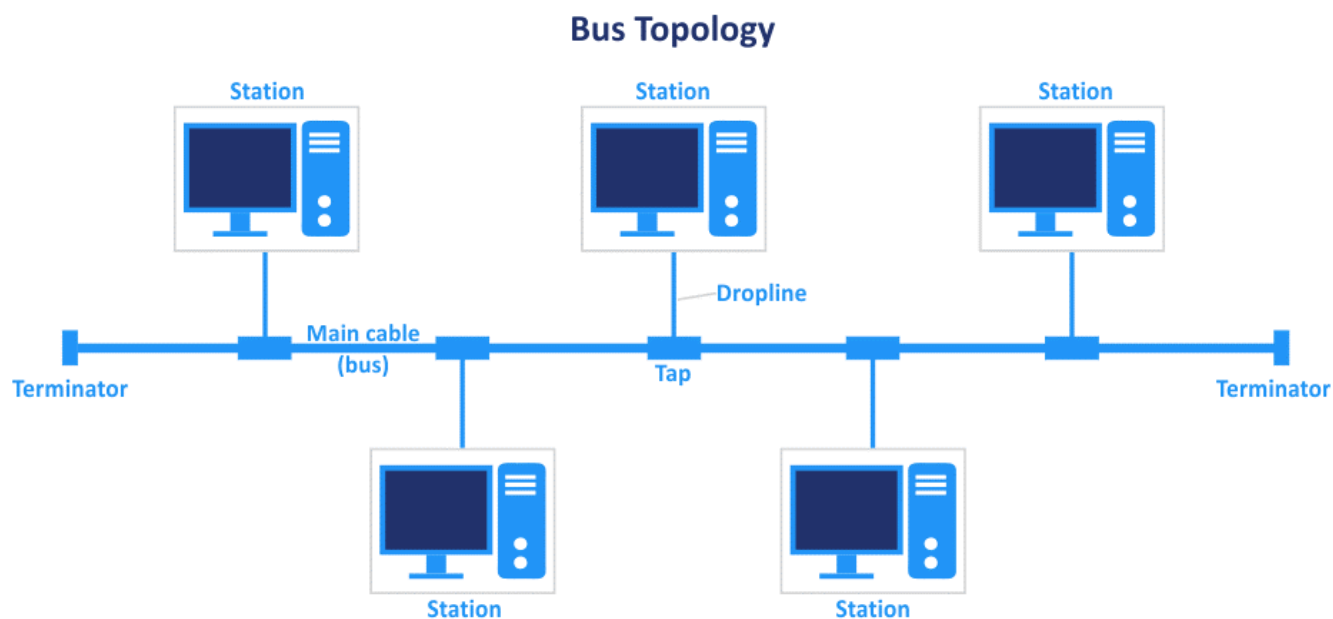


Figure 2: Bus Topology Structure

2. Detailed Comparative Analysis

Comparison Parameter	Star Topology	Bus Topology
Initial Cost	Higher because a central switch/hub and individual cables are required	Lower because a single backbone cable is shared
Cabling Requirement	Requires separate cable connections from each device to the central device	Requires one main backbone cable
Installation	Relatively easy to install and configure	Simple to install because of fewer connections
Reliability	High; failure of one node or its cable normally affects only that node	Lower; failure of the backbone can affect the entire network
Performance	Better performance because communication is managed through the central device	Performance decreases as more devices share the common medium
Data Collisions	Lower in switched networks	Higher possibility because devices share the transmission medium
Fault Detection	Easier to identify and isolate individual device or cable faults	Difficult to locate faults in the shared backbone
Scalability	Can be expanded by adding devices or additional networking equipment, subject to switch capacity	Limited because additional devices increase traffic on the shared backbone
Maintenance	Easier because individual connections can be checked separately	More difficult because backbone problems can affect many devices
Failure Impact	Failure of one node generally does not affect other nodes; central-device failure can affect the network	Backbone failure can bring down the entire network

Suitability	Suitable for modern LANs and organizational networks	Suitable mainly for small or simple networks
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3. Cost Analysis

Bus topology has a cost advantage because all devices use a common backbone cable. It requires fewer cables and fewer networking components, making the initial installation relatively inexpensive.

Star topology has a higher initial cost because every device requires an individual connection to a central switch or hub. However, the additional cost provides better control, easier troubleshooting, and improved network performance.

Therefore, **Bus topology is preferable when minimizing initial installation cost is the primary objective**, while Star topology is more suitable when reliability and performance are more important.

4. Reliability Analysis

Star topology provides better reliability at the individual-device level. If the cable connecting one computer to the switch fails, the other computers can normally continue communicating.

In Bus topology, all devices depend on the same backbone. Therefore, a failure in the main cable can interrupt communication for the entire network.

Hence, **Star topology provides better reliability for networks where continuous connectivity is important**.

5. Performance Analysis

Star topology generally provides better performance because communication is managed through the central networking device. In modern switched networks, devices can communicate using dedicated switch ports, reducing unnecessary traffic and collisions.

Bus topology uses a shared communication medium. As the number of connected devices increases, more devices compete for the same transmission medium. This can result in increased traffic and reduced network performance.

Thus, **Star topology is more suitable for networks with many active devices and higher traffic requirements**.

6. Practical Example

Consider a **college computer laboratory with 30 computers**.

If Star topology is used, every computer can be connected to a central switch. If one computer's network cable becomes faulty, the remaining computers can continue operating. The faulty connection can also be identified and repaired more easily.

If the same laboratory used Bus topology, all computers would share a common backbone. Increasing the number of computers would increase traffic on the shared medium, and a failure of the backbone could disrupt communication throughout the laboratory.

Therefore, for a college laboratory, **Star topology would be the more practical choice** because of its better reliability, performance, and maintainability.

7. Analytical Comparison

The choice between Star and Bus topology depends on the requirements of the network.

- If **low cost and simple installation** are the main priorities, Bus topology has an advantage.
- If **reliability and fault isolation** are important, Star topology is preferable.
- For **higher network traffic**, Star topology provides better performance.
- For **larger or expandable networks**, Star topology is more practical.
- For **easy maintenance and troubleshooting**, Star topology is superior.
- Bus topology can be considered for **small and simple networks** where cost is a major concern.

This shows that no topology is universally best; the appropriate topology depends on the network's size, budget, traffic requirements, reliability requirements, and maintenance needs.

Conclusion

Star and Bus topologies differ considerably in their architecture and operational characteristics. **Bus topology is economical and simple to install because it uses a shared backbone cable**, making it suitable for small and simple networks. However, its shared medium creates limitations in reliability and performance as the number of devices increases.

Star topology requires greater initial investment because of the central switch/hub and individual cables, but it provides better reliability, performance, fault isolation, and maintenance. Therefore, for modern organizational networks, laboratories, offices, and other environments where performance and reliability are important, **Star topology is generally the more suitable choice**. The assessment material similarly identifies Star as superior in reliability and performance, while Bus has advantages in cost and simplicity.

Q2. Comparative Analysis of Mesh and Star Topologies

Introduction

Mesh topology and **Star topology** are network topologies that differ significantly in their connection structure and fault tolerance. In a Mesh topology, network devices are interconnected through multiple links, creating alternative paths for data transmission. In a Star topology, all devices are connected to a central networking device such as a switch or hub.

The major factor considered in this comparison is **fault tolerance and reliability**, because the availability of alternate communication paths directly affects the ability of a network to continue operating when a connection fails. The assessment material identifies Mesh topology as highly fault tolerant, while Star topology provides moderate fault tolerance because it depends on the central device.

1. Structural Comparison

Mesh Topology

In a Mesh topology, devices are connected to multiple other devices. In a full mesh, every device has a direct connection with every other device.

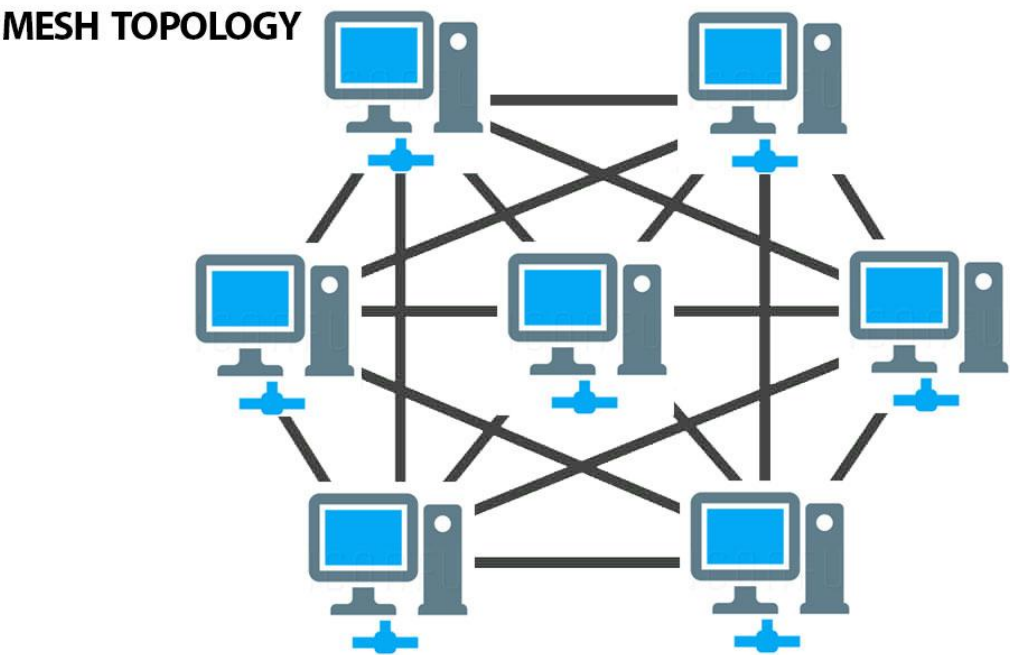


Figure 3: Mesh Topology Structure

Star Topology

In Star topology, each device has an individual connection to a central switch or hub.

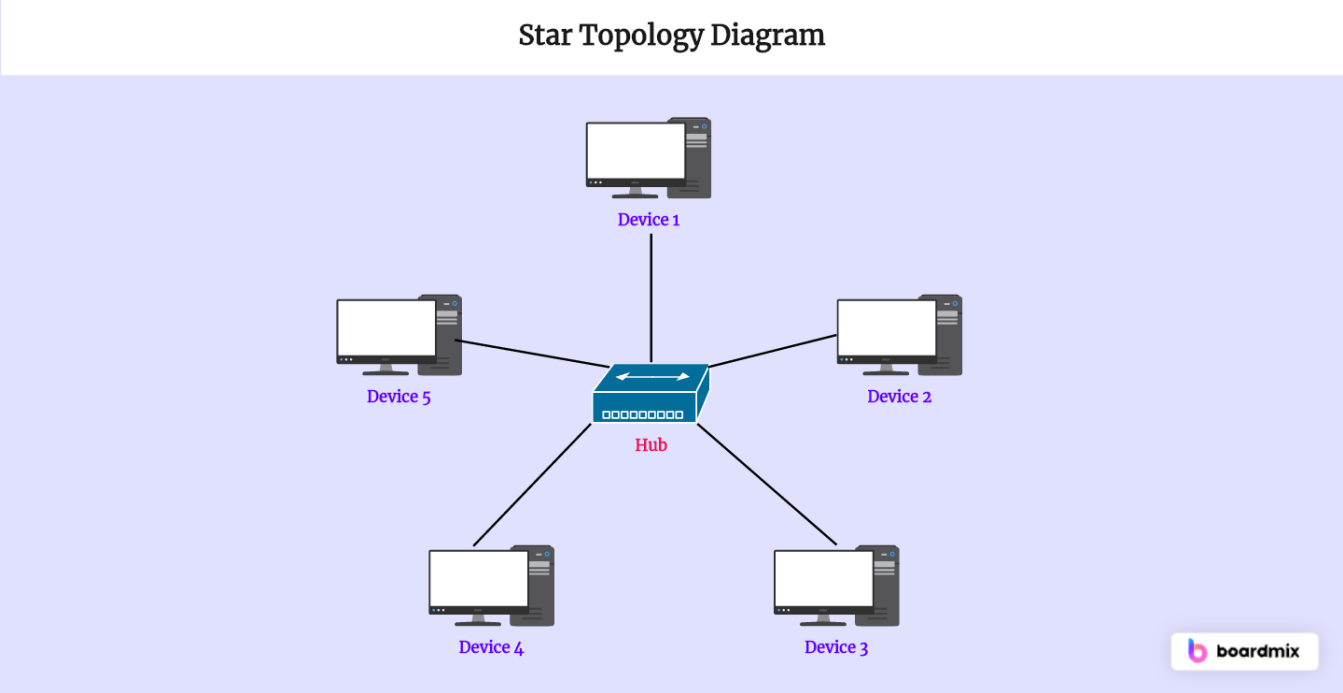


Figure 4: Star Topology Structure

2. Detailed Comparative Analysis

Comparison Parameter	Mesh Topology	Star Topology
Connection Structure	Devices have multiple interconnections	Devices connect to a central device
Fault Tolerance	Very high	Moderate
Alternative Paths	Multiple alternative paths are available	Generally depends on the central device
Reliability	Extremely high	High for individual node failures
Failure of One Link	Communication can continue through another path	The affected device may lose connectivity
Central Device Dependency	No single central device controls the entire network	Highly dependent on the central switch/hub
Central Device Failure	No equivalent single point of failure in a full mesh	Can affect the entire network
Cost	High because of numerous links and ports	Lower than full mesh
Installation Complexity	Very high	Relatively simple
Maintenance	Difficult because of many connections	Easier to troubleshoot
Scalability	Difficult and expensive as devices increase	Easier to expand, subject to switch capacity
Suitability	Critical environments requiring high availability	General LANs and organizational networks

3. Fault Tolerance Analysis

The major advantage of Mesh topology is its **high fault tolerance**. Since devices have multiple connections, failure of one communication link does not necessarily interrupt the entire network. Data can be redirected through an alternative path.

For example, if a direct connection between Device A and Device B fails, the network may route communication through another device.

In contrast, Star topology has **moderate fault tolerance**. Failure of a single device or its connecting cable normally affects only that particular device. However, the central switch or hub represents a critical point. If the central device fails, communication between the connected devices can be disrupted across the network.

4. Reliability Analysis

Mesh topology provides greater reliability because of its redundant connections. Multiple paths allow the network to continue functioning even when some links fail.

Star topology is also reliable for individual connection failures. For example, if one computer's cable fails, other computers can generally continue communicating through the switch.

However, the central switch or hub becomes a **single critical point of failure**. Therefore, Mesh topology is more appropriate where maximum availability and fault tolerance are required.

5. Cost and Complexity Analysis

Mesh topology requires a large number of cables, ports, and connections. As the number of devices increases, the number of required connections grows rapidly. This makes installation expensive and technically complex.

Star topology requires individual cables from each device to the central switch or hub. Although this requires more cabling than a simple Bus topology, it is significantly easier to implement and manage than a full Mesh topology.

Therefore, **Star topology provides a better balance between cost, reliability, and manageability**, whereas Mesh topology prioritizes maximum fault tolerance.

6. Maintenance and Troubleshooting

Mesh topology is difficult to maintain because it contains a large number of connections. Identifying and troubleshooting a faulty link can require checking several possible communication paths.

Star topology is comparatively easier to maintain. Since each device has a separate connection to the central device, a network administrator can isolate a faulty device or cable more easily.

Therefore, **Star topology is more practical for networks where simple maintenance and troubleshooting are important**.

7. Practical Example

Consider a **critical communication network in an environment where continuous connectivity is essential**. A Mesh topology would be appropriate because multiple communication paths are available. If one connection fails, another path can be used to maintain communication.

For a **college laboratory or office LAN**, Star topology would generally be more practical. Individual device failures can be isolated, while the central switch provides an organized and manageable network structure.

Thus, the choice depends on whether the priority is **maximum fault tolerance** or **cost-effective and manageable networking**.

8. Analytical Comparison

The comparison demonstrates that Mesh topology is superior when **fault tolerance and maximum reliability** are the primary requirements. Its redundant connections allow communication to continue even when individual links fail. However, this reliability comes at the cost of increased **installation complexity, cabling requirements, hardware requirements, and maintenance effort**. Star topology provides a practical compromise. It offers good reliability for individual node failures while being easier to install, manage, troubleshoot, and expand. Its main limitation is dependency on the central switch or hub.

Therefore:

- **Maximum fault tolerance → Mesh**
- **Lower complexity → Star**
- **Easier maintenance → Star**
- **Multiple alternative paths → Mesh**
- **General organizational LAN → Star**
- **Critical/high-availability environment → Mesh**

Conclusion

Mesh and Star topologies provide different levels of reliability and fault tolerance. **Mesh topology is highly robust because multiple communication paths are available, allowing the network to continue operating when individual links fail**. However, its large number of connections makes it expensive and complex to install and maintain.

Star topology provides moderate fault tolerance and is easier to manage, but its central switch or hub can become a single point of failure. Therefore, Mesh topology is more suitable for **critical environments where maximum reliability and fault tolerance are required**, whereas Star topology is generally more suitable for **cost-effective, manageable, and scalable organizational networks**. This conclusion is consistent with the assessment material's comparison of Mesh's high fault tolerance against Star's moderate fault tolerance.

Q3. Comparative Analysis of Bus and Mesh Topologies

Introduction

Bus topology and **Mesh topology** differ significantly in terms of installation complexity, network structure, cabling requirements, cost, and scalability. Bus topology uses a **single backbone cable** to connect multiple devices, whereas Mesh topology uses **multiple direct connections between devices**.

The main advantage of Bus topology is its simplicity. Mesh topology, in contrast, provides a highly interconnected structure but requires considerably more cables and connections. Therefore, Bus topology is easier and less expensive to install, while Mesh topology is more complex and costly.

1. Structural Comparison

Bus Topology

In Bus topology, all devices are connected to a single shared backbone cable.

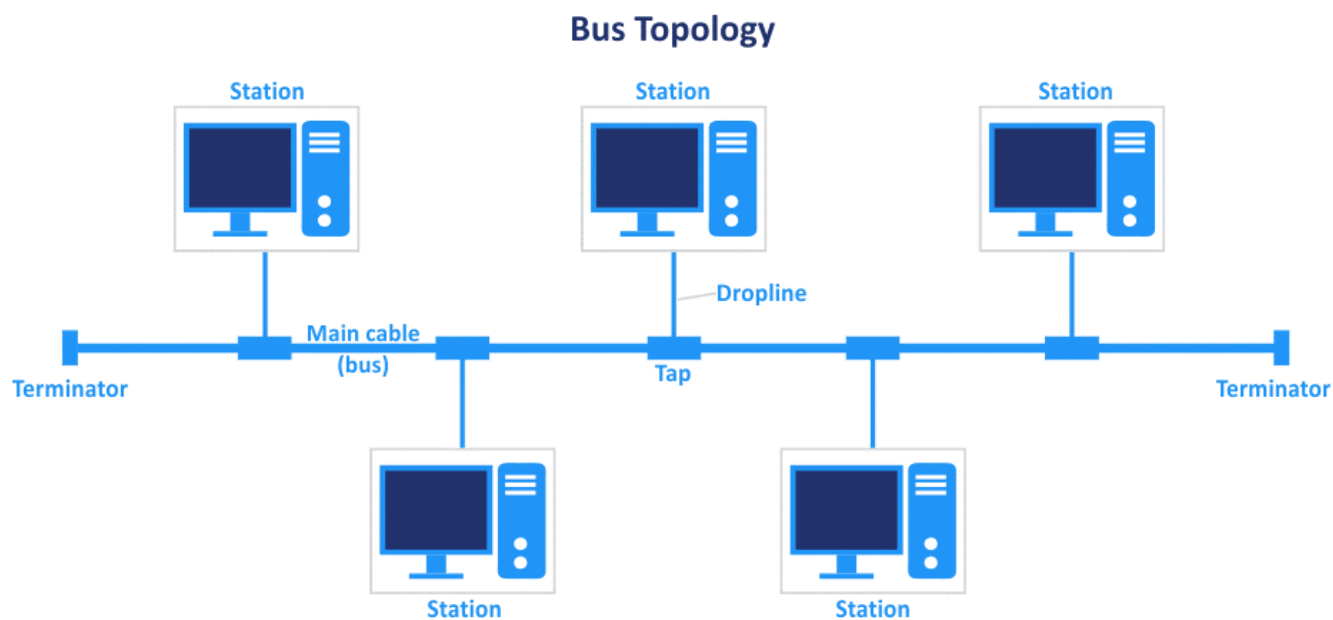


Figure 5: Bus Topology Structure

Mesh Topology

In Mesh topology, devices are connected through multiple links. In a full Mesh topology, each device has a direct connection with every other device.

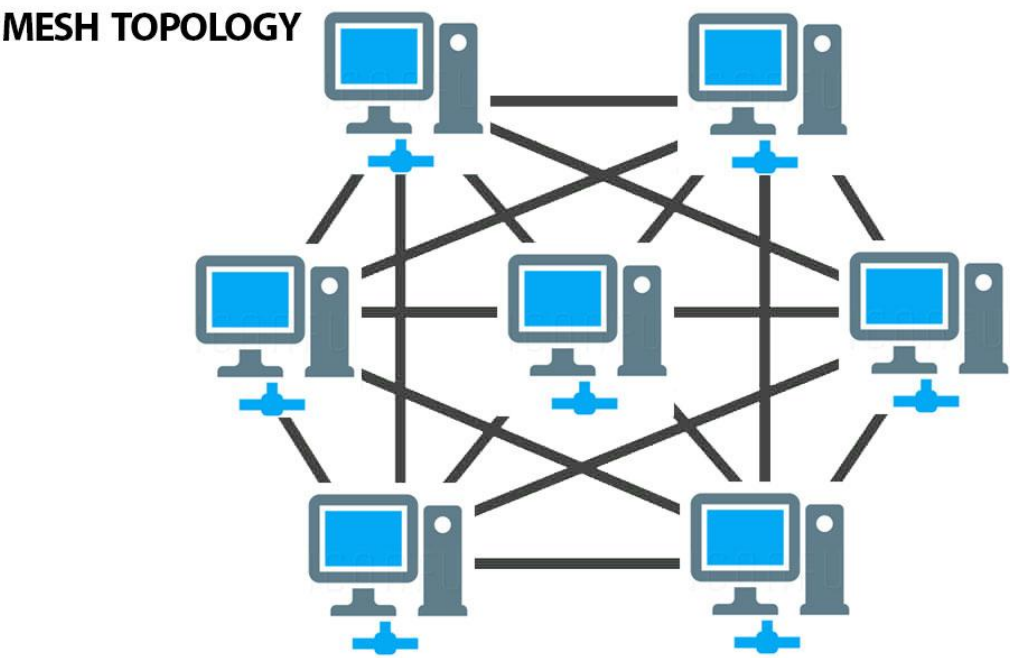


Figure 5: Mesh Topology Structure

2. Detailed Comparative Analysis

Comparison Parameter	Bus Topology	Mesh Topology
Network Structure	Single shared backbone	Multiple interconnected links
Installation Complexity	Low	Very high
Cabling Requirement	Minimal	Very high
Initial Cost	Low	High
Number of Connections	Relatively few	Large number of connections
Configuration	Simple	Complex
Maintenance	Relatively simple for small networks but backbone faults are difficult to locate	Complex because many links must be monitored
Scalability	Limited	Difficult and expensive as devices increase
Fault Tolerance	Lower	Very high
Alternative Paths	Limited	Multiple paths available
Installation Time	Shorter	Longer
Suitable Environment	Small/simple networks	Critical and highly reliable networks

3. Installation Complexity

Bus topology is relatively easy to install because all devices share a common backbone cable. It requires fewer physical connections, making the setup straightforward.

Mesh topology is much more complicated because devices require multiple connections. In a full Mesh topology, every device needs a direct connection with every other device.

As the number of devices increases, the number of required links increases rapidly. Consequently, installation becomes more time-consuming and technically demanding.

4. Cabling Requirements

One of the major differences between the two topologies is their cabling requirement.

Bus topology uses a common backbone cable. Therefore, fewer cables and networking connections are required.

Mesh topology requires numerous point-to-point connections. Each additional device can require several additional links to other devices.

For this reason, Mesh topology requires considerably more cabling than Bus topology.

5. Cost Analysis

Bus topology is generally more economical because it requires fewer cables and connections. This makes it suitable for small networks where minimizing installation cost is important.

Mesh topology has a much higher implementation cost because of its large number of cables, ports, and connections. The additional hardware and installation effort increase the overall cost.

Therefore, **Bus topology has a clear cost advantage over Mesh topology.**

6. Maintenance and Troubleshooting

Bus topology has a relatively simple structure, which can make basic maintenance straightforward. However, the backbone cable is critical to the entire network. If the backbone develops a fault, multiple or all devices may lose connectivity, and locating the exact fault can be difficult.

Mesh topology is more complicated to maintain because of its large number of interconnected links. Network administrators may need to monitor and troubleshoot many individual connections.

Therefore, although Mesh provides greater reliability, its **maintenance requirements are higher because of its complex structure.**

7. Scalability Analysis

Bus topology has limited scalability. As more devices are connected, traffic on the shared backbone increases. The network can therefore become less efficient as it grows.

Mesh topology can technically accommodate additional devices, but expansion becomes increasingly difficult and expensive because new devices need multiple connections to existing devices.

Thus, both topologies face scalability challenges, but **Mesh topology has greater implementation complexity when the network size increases.**

8. Practical Example

Consider a **small temporary computer network** containing a limited number of devices.

Bus topology may be suitable because it requires minimal cabling and can be installed relatively quickly. The simple structure makes it appropriate when the network does not require extensive redundancy.

In contrast, consider a **critical communication environment** where continuous connectivity is essential. Mesh topology would be more appropriate because multiple communication paths can provide redundancy if one link fails.

Therefore, the appropriate topology depends on whether **simplicity and low cost** or **reliability and redundancy** are the primary requirements.

9. Analytical Comparison

The comparison shows a clear trade-off between **simplicity and reliability**.

Bus topology has a simple structure with minimal cabling, making it easier and less expensive to install. However, its shared backbone creates limitations in terms of reliability and scalability.

Mesh topology has a complex structure requiring many connections. This increases installation time, cost, and maintenance requirements. However, the same interconnected structure provides multiple communication paths and significantly improves fault tolerance.

Therefore:

- **Simple installation → Bus**
- **Lower cabling requirement → Bus**
- **Lower initial cost → Bus**
- **High redundancy → Mesh**
- **Multiple communication paths → Mesh**
- **Higher fault tolerance → Mesh**
- **Simple small network → Bus**
- **Critical/high-reliability network → Mesh**

Conclusion

Bus and Mesh topologies represent two different approaches to network design. **Bus topology is simple, economical, and requires minimal cabling**, making it suitable for small and straightforward networks. However, its shared backbone creates limitations in fault tolerance and scalability.

Mesh topology provides superior redundancy and reliability through multiple interconnected paths, but this comes with significantly greater installation complexity, cabling requirements, cost, and maintenance effort. Therefore, **Bus topology is preferable when simplicity and low cost are the main priorities, while Mesh topology is preferable when reliability and fault tolerance are more important than installation complexity and cost.**

Q4. Comparative Analysis of Hybrid and Star Topologies

Introduction

Hybrid topology and **Star topology** are network structures used to connect multiple devices and network segments. A **Star topology** connects all devices to a central switch or hub, whereas a **Hybrid topology** combines two or more different network topologies, such as Star, Bus, or Mesh, according to the requirements of the organization.

The key difference between them is **scalability and flexibility**. Hybrid topology can combine different network structures and allows organizations to expand their networks according to their specific requirements. Star topology is also scalable, but its expansion is influenced by the capacity and available ports of the central networking device.

1. Structural Comparison

Star Topology

In Star topology, every device is connected individually to a central switch or hub.

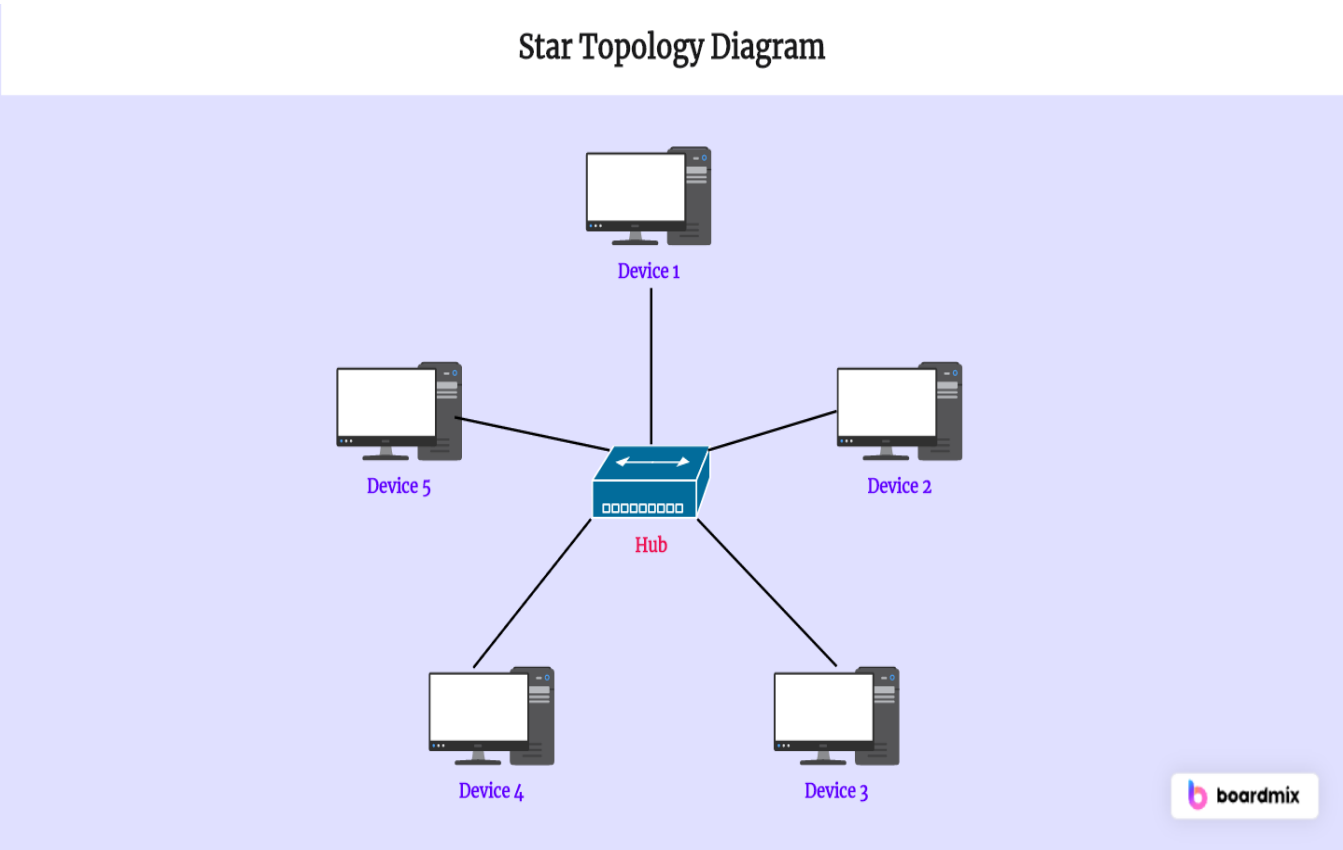


Figure 7: Star Topology Structure

Hybrid Topology

Hybrid topology combines two or more different topologies to create a network structure suitable for organizational requirements.

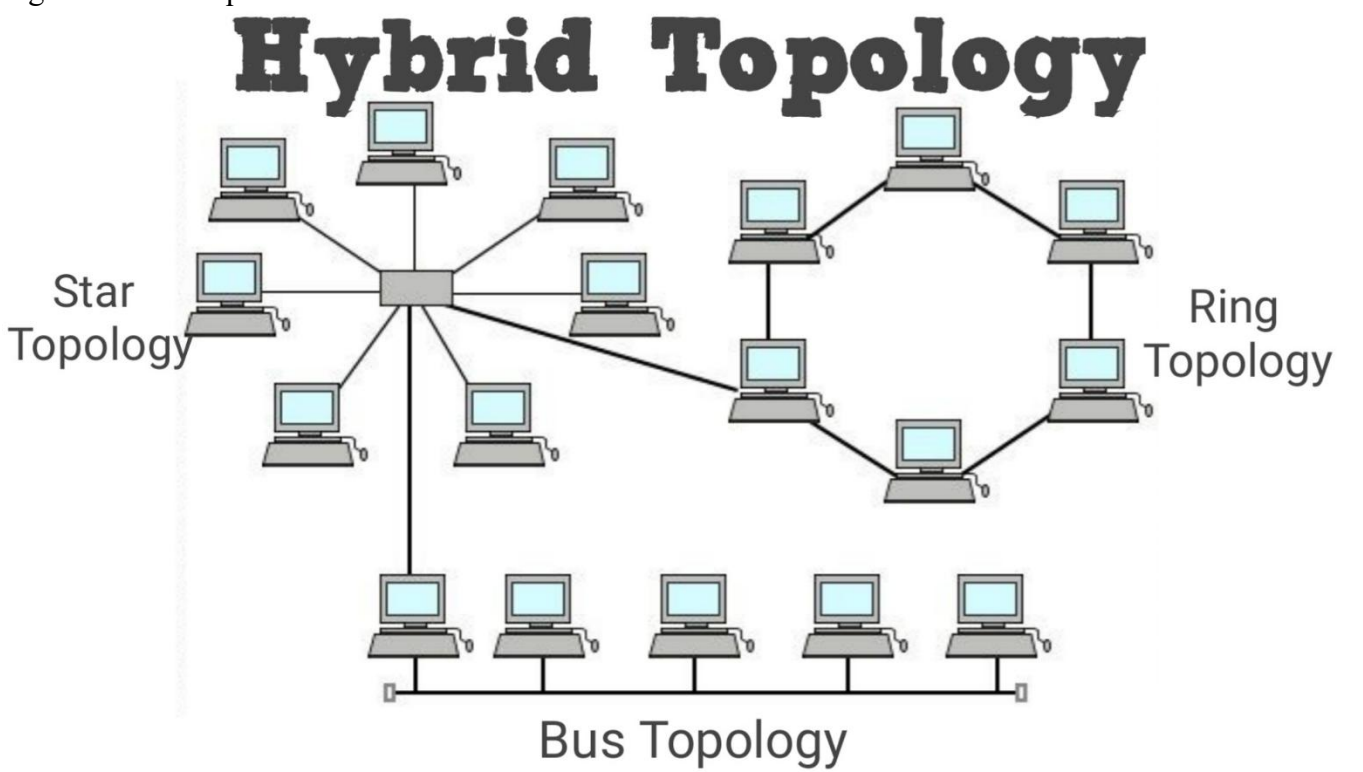


Figure 8: Hybrid Topology Structure

2. Detailed Comparative Analysis

Comparison Parameter	Hybrid Topology	Star Topology
Network Structure	Combination of two or more topologies	Devices connected to one central device
Flexibility	Very high	Moderate
Scalability	High	Moderate to high
Expansion	New network segments can be added flexibly	Expansion depends on central device capacity
Installation Complexity	High	Relatively simple
Cost	Generally higher	Relatively lower
Maintenance	Moderate to complex	Easier
Network Management	Requires management of different topology segments	Comparatively simple
Fault Isolation	Depends on the topology used in each segment	Individual node faults can be isolated
Central Device Dependency	Depends on the design	High
Suitability	Large and complex organizations	Small to medium LANs and structured networks

3. Scalability Analysis

The major advantage of Hybrid topology is its **high scalability**. Since it combines different topologies, organizations can expand individual sections according to their requirements.

For example, an organization may use Star topology within individual departments and connect those departments using another suitable topology. A new department can then be added as a separate network segment without completely redesigning the existing network.

Star topology also supports expansion, but the number of devices that can be directly connected depends on the available ports and capacity of the central switch or hub. Once the central device reaches its capacity, additional networking hardware may be required.

4. Flexibility Analysis

Hybrid topology provides greater flexibility because different topology types can be selected for different parts of the network.

For example:

- A **Star topology** can be used inside an office department.
- A **Mesh segment** can be used where high reliability is required.
- A **Bus segment** can be used for a simple network section.

This allows the overall network to be designed according to the requirements of different departments or locations.

Star topology has a more uniform structure because all devices connect to the central switch or hub. This makes it simpler but less flexible than a Hybrid topology.

5. Cost and Installation Analysis

Star topology is generally easier to install because its structure is straightforward. Each device is connected to a central networking device, making configuration and physical deployment relatively simple.

Hybrid topology can be more expensive and complex because it may require different networking devices, cabling arrangements, and configuration methods for different network segments.

Therefore, **Star topology is preferable when simplicity and controlled installation costs are important**, whereas Hybrid topology is more appropriate when flexibility and customized network design justify the additional complexity.

6. Maintenance Analysis

Star topology is relatively easy to maintain because each device has an individual connection to the central device. If a particular cable or device fails, the problem can generally be isolated to that connection.

Hybrid topology has moderate to high maintenance requirements because different sections may use different topology structures. Network administrators need to understand and monitor each component and its associated connections.

The assessment material specifically notes that Hybrid topology requires careful monitoring and skilled administration because multiple topology types are combined.

7. Reliability and Fault Isolation

Star topology provides good fault isolation for individual devices. If one computer or its connecting cable fails, other devices can normally continue operating.

However, failure of the central switch or hub can affect the entire Star network.

Hybrid topology can be designed to provide different levels of reliability in different sections. For example, a Mesh segment can be incorporated where high fault tolerance is required, while Star segments can be used in ordinary departments.

Thus, **Hybrid topology can provide a more customized reliability structure**, depending on the topologies selected.

8. Practical Example

Consider a **large university campus** containing multiple departments such as Computer Science, Administration, Library, and Research laboratories.

Each department could use a Star topology internally because it is easy to manage. The different departments can then be interconnected using a suitable combination of networking structures.

This creates a Hybrid topology that can accommodate the different requirements of each department.

In comparison, using a single Star topology for the entire campus would make the network dependent on central networking infrastructure and could require additional switches when the number of devices increases.

Therefore, **Hybrid topology is more suitable for a large and diverse organization with different networking requirements.**

9. Analytical Comparison

The comparison demonstrates that Hybrid topology provides greater **flexibility and scalability**, while Star topology provides greater **simplicity and ease of management**.

Hybrid topology is particularly useful when an organization has different departments, locations, or applications requiring different networking characteristics. It allows the network designer to combine appropriate topologies instead of using one structure throughout the entire network.

Star topology is preferable when the network has a relatively straightforward structure and centralized management is desired.

Therefore:

- **Higher flexibility → Hybrid**
- **Higher scalability → Hybrid**
- **Simple installation → Star**
- **Easy maintenance → Star**
- **Customized network design → Hybrid**
- **Simple LAN structure → Star**
- **Large organizational network → Hybrid**

Conclusion

Hybrid and Star topologies provide different advantages for network design. **Star topology is simple, organized, and relatively easy to install and maintain**, making it suitable for conventional LAN environments. However, its expansion depends on the capacity of the central networking device.

Hybrid topology provides greater scalability and flexibility because it combines multiple topology types and allows new network segments to be added according to organizational requirements. Although it requires more careful planning, monitoring, and skilled administration, it is more suitable for **large, complex, and expanding organizations**. Therefore, when flexibility and long-term scalability are the primary requirements, **Hybrid topology is the better choice**, while Star topology is preferable when simplicity and ease of management are the priorities.

Q5. Comparative Analysis of Star, Mesh, Bus, and Hybrid Topologies with Respect to Maintenance

Introduction

Network maintenance refers to the activities involved in **monitoring, troubleshooting, identifying faults, repairing connections, and maintaining network performance**. The maintenance requirements of a network depend greatly on its topology.

Among the four topologies, **Star topology is relatively easy to maintain**, while **Mesh topology is more difficult because of its large number of connections**. Bus topology presents difficulties when faults occur in the backbone cable, whereas Hybrid topology has moderate maintenance complexity because it combines multiple topology structures.

1. Maintenance Characteristics of Different Topologies

Parameter	Star Topology	Mesh Topology	Bus Topology	Hybrid Topology
Maintenance Complexity	Low	Very High	Moderate	Moderate to High
Fault Identification	Easy	Difficult	Difficult	Depends on design
Troubleshooting	Relatively easy	Time-consuming	Difficult for backbone faults	Requires skilled administration
Cable Management	Easy to organize	Complex	Simple structure	Complex
Fault Isolation	Easy for individual nodes	Possible through multiple paths but many links must be checked	Difficult when backbone fails	Depends on individual segments
Network Monitoring	Relatively simple	Complex	Relatively simple but backbone-critical	Requires monitoring of multiple topologies
Maintenance Cost	Relatively low	High	Low to moderate	Moderate to high
Administrator Skill Requirement	Moderate	High	Moderate	High
Suitable Environment	Offices, laboratories, LANs	Critical/high-reliability networks	Small/simple networks	Large and complex organizations

2. Star Topology – Maintenance

Star topology is generally **easy to maintain and troubleshoot** because each device has an individual connection to a central switch or hub.

If one computer stops communicating, the administrator can inspect that particular device, cable, or switch port without necessarily disturbing the other devices.

For example, if Computer 5 loses network connectivity, the administrator can check:

1. Computer 5's network interface.
2. Its individual cable.
3. The corresponding switch port.
4. The network configuration.

This makes fault identification relatively quick and localized.

Therefore, **Star topology is highly suitable when easy maintenance and quick troubleshooting are important.**

3. Mesh Topology – Maintenance

Mesh topology has **high maintenance complexity** because it contains a large number of connections between devices.

Although its multiple links provide excellent reliability, they also increase the number of components that must be monitored and maintained. When a fault occurs, the administrator may need to inspect multiple links and communication paths.

For example, in a network containing several interconnected devices, identifying a faulty connection requires checking the relevant links and determining whether traffic is being successfully routed through alternative paths.

Therefore, Mesh topology provides excellent reliability but requires **more time, technical expertise, and maintenance effort.**

4. Bus Topology – Maintenance

Bus topology has a relatively simple physical structure because all devices use a common backbone cable. However, this shared backbone becomes a major challenge during troubleshooting.

If the backbone cable develops a fault, communication for multiple devices can be disrupted. Locating the exact position of the fault along the backbone can be difficult and time-consuming.

Therefore, although Bus topology is simple in structure, **backbone failures can make maintenance difficult.**

5. Hybrid Topology – Maintenance

Hybrid topology combines two or more different topologies. Its maintenance complexity therefore depends on the topologies included in the network.

For example, a network may contain Star topology in one department and Mesh topology in another. The administrator must understand and manage both structures.

This provides flexibility but also requires **careful monitoring and skilled administration**. Troubleshooting may involve checking different network segments and their associated devices and connections.

6. Comparison of Fault Identification

Star

Faults can generally be isolated to an individual device, cable, or switch port.

Result: Easy fault identification.

Mesh

Multiple connections and alternative paths make the network robust, but identifying a particular faulty link can require checking several connections.

Result: More complex fault identification.

Bus

A fault in the backbone can affect many devices, making the location of the problem difficult to determine.

Result: Difficult backbone troubleshooting.

Hybrid

Fault identification depends on which topology segment contains the problem.

Result: Moderate to high complexity.

7. Maintenance Cost Analysis

Star topology generally has a lower maintenance burden because faults can be isolated and repaired individually.

Mesh topology has higher maintenance costs because its large number of connections requires more monitoring, testing, and technical expertise.

Bus topology may have lower routine maintenance requirements, but troubleshooting a backbone failure can require significant effort.

Hybrid topology can have moderate to high maintenance costs because administrators must maintain multiple topology types and their associated networking components.

Thus, **maintenance cost is influenced not only by the number of devices but also by the complexity of the network structure.**

8. Practical Example

Consider a **large corporate organization** with several departments.

If Star topology is used within each department, network administrators can easily identify problems affecting individual computers. A faulty cable can be replaced without restructuring the entire network.

If Mesh topology is used for critical communication links, it provides greater reliability but requires specialized personnel to monitor the numerous connections.

If Bus topology is used, a backbone failure could affect many devices simultaneously.

If Hybrid topology is implemented, different departments can use different topologies according to their requirements. However, administrators must have sufficient knowledge to maintain each network segment.

Therefore, for an organization, **Star topology is generally easier to maintain**, while Hybrid topology may be selected when flexibility and different departmental requirements are more important.

9. Analytical Comparison

The maintenance comparison demonstrates an important trade-off between **simplicity, reliability, and flexibility**.

- **Star topology** provides the easiest maintenance because individual connections can be isolated.
- **Mesh topology** provides excellent reliability but requires greater maintenance effort because of its complex connections.
- **Bus topology** has a simple structure but backbone failures can be difficult to locate.
- **Hybrid topology** provides flexibility but requires skilled administrators because multiple topology types must be managed.

Therefore, selecting a topology should not be based only on installation cost or performance. **Maintenance requirements should also be considered because a network that is difficult to troubleshoot can increase downtime and operational costs.**

10. Overall Comparison

Topology	Main Maintenance Advantage	Main Maintenance Limitation
Star	Easy fault isolation and troubleshooting	Central switch/hub is important to the network
Mesh	Multiple paths provide resilience	Large number of connections makes maintenance complex
Bus	Simple physical structure	Backbone faults are difficult to locate
Hybrid	Flexible maintenance according to network segments	Requires skilled administration and careful monitoring

Conclusion

Maintenance requirements vary significantly among Star, Mesh, Bus, and Hybrid topologies. **Star topology is generally the easiest to maintain because faults can be quickly identified and isolated to individual devices or cables.** Mesh topology provides excellent reliability but requires greater maintenance effort because of its complex network of connections.

Bus topology is simple but presents significant troubleshooting challenges when the backbone cable fails. Hybrid topology offers flexibility and scalability but requires careful monitoring and skilled administration because it combines multiple topology structures.

Therefore, **Star topology is the most suitable when easy maintenance and troubleshooting are the primary requirements**, while Mesh and Hybrid topologies are better suited to environments where **high reliability, flexibility, or specialized network requirements justify greater maintenance complexity.**

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured

Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion